

VENKATA YASHWANT KUMAR BHYRI

(+86) 15914122353 | 123040005@link.cuhk.edu.cn | [LinkedIn](#) | 518172, Longgang District, Shenzhen, China

EDUCATION:

The Chinese University of Hong Kong, Shenzhen (*B.Eng. in Computer Science and Engineering*)

2023 – 27

- **Full Admission Excellence Scholarship** – School of Data science, CUHK-SZ
- **Guangdong Government Outstanding International Student Scholarship**
- **2025 Leading Academic Peer Advisor @** School of Data Science, CUHK-SZ

EXPERIENCE:

AI Agent Development Engineer Intern - R&D, AIGC Algorithms - Wondershare Filmora @ Shenzhen, China 2026 Jan - July

- Built a **multi-agent AIGC workflow** orchestration layer, transforming a single media generation request into a staged multimodal production graph spanning video generation, custom AI-music integration, dialogue/TTS, captions and visual effects.
- Prototyped a dedicated **trend research agent** with **memory-retrieval** and **ML** recommendation layers that converted **700+** real-time TikTok/Douyin-style creative trend signals : viral audio patterns, popular themes/styles, script hooks, color palettes, transition tempo, and effects into reusable context packets, skill files, design documents, and FX native editing parameters.
- Designed the agent runtime with strong **reasoning**, planning, **memory retrieval**, and prompt versioning with human-in-the-loop approval gates. Built a semantic **prompt harness/context compiler** that merges user intent, trend-agent recommendations, multi-agent outputs, and retrieved context into executable **tool-calling** instructions for automated media generation at scale.
- Built **A/B workflow observability** and **RL**-style agent evaluation feedback loop using *OpenTelemetry*-style traces, metrics, agent evals, tool-call traces, regression tests, guardrail events, latency/error metrics, and API-cost attribution; leading to **API cost** optimization by **28%** and reduced **orchestration latency** to **2%** of E2E generation latency across long-running AIGC workflows.

AI Model Engineer : Audio Algorithm Developer - Optek Microelectronics @ Shenzhen, China

2025 July – Sept

- Engineered a **full-stack TinyML Audio** Classification Pipeline integrating Digital Signal Processing (**DSP**) for real-time feature extraction (log-Mel spectrograms, **MFCCs**), developed custom **2-layer fine-tuned** network architecture on top of Google-YAMNet; optimized numerically stable SoftMax functioning for robust training data convergence and gradient stability.
- Implemented a Linux-based embedded deployment workflow by containerizing the model with Docker, **TensorFlow** Lite-Micro and packaged for **SDK** integration. Achieved **93%+** prediction accuracy (industry-standard) with **INT8 quantization**.
- Optimized model performance for a 700 MHz DSP + 16 MB custom **SoC NPU (XNCC environment)**, achieving **<10 ms** latency and **14×** model compression via **INT8 quantization**, **pruning**, and C++ operator tuning.

Research Assistant : Multimodal LLM's - BIRD-SQL [HKU x Google Cloud]

2025 Jun – Sept

- Reconstructed an advanced **multi-modal** benchmark framework for Visual Question Answering (**VQA**) on BIRD-SQL dataset (12,751+ question-SQL pairs) by integrating **OCR-driven** Context Grounding (ResNet) to augment **Llama/GPT-4's** reasoning for complex DB-queries in Text-to-SQL (Hybrid SQL) generation. [**Vision + NLP**]
- Designed **DSPy**-optimized Schema-aware Prompting strategies & Complex SQL architecture to evaluate User Intent in LangGraph based **RAG** workflows (**FAISS**-indexed). Implemented hybrid SQL queries to mitigate hallucinations. Built Execution Accuracy (**EX**) protocols to diagnose Chain-of-Thought (**CoT**) reasoning failures across **95+** large-scale databases

Research Assistant - NLP Benchmark Optimization for SLM Distillation @ CUHK-SZ Natural Language Processing Group 2024

- Automated critical Python **SLM Distillation** Pipelines (**HF Datasets – Pandas – Scikit-learn**) using Hugging Face **TRL**. Deployed “LLM-as-Judge” framework to filter hallucination in synthetic training data, reduced factual errors by **31%** in **SLM** outputs
- Engineered a large-scale evaluation framework via **TRL's Judge API** to benchmark for **200+** LLM response pairs, developing interactive **Streamlit** dashboards to visualize **BERT Score/ROUGE** metric to boost Truth-layer Alignment accuracy by **25%**

PROJECTS: [Link to personal portfolio website](#)

1. Antigravity : AI-Native Technical Interviewing Platform

Next.js, React, TypeScript, FastAPI, Redis, WebRTC/WebSocket, ASR/STT, TTS, LLM APIs, RBAC, Docker, Playwright

2. MindScape : Multimodal Psychiatric Clinical Intelligence System

ASR/event detection, Emotion2Vec+ modelling, MedCPT embeddings, GMU fusion, HNSW, BM25, RRF, BioLinkBERT reranking, RAG - grounded LLM-CoT, DeBERTa-v3 NLI, DSM-5 rule engine, Docker/Kubernetes, OAuth2/JWT

3. Lalamove AI-Ops R&D Project: Logistics AI Ops Platform

Next.js, React, TypeScript, Python/FastAPI, Kafka, Flink, Redis, PostgreSQL, ClickHouse, S3 Data Lake, dbt, Airflow, WebSockets, H3 Geospatial Indexing, OR-Tools, XGBoost, Vector Search (Milvus), Hybrid Retrieval (BM25 + HNSW + RRF), Agentic RAG, Long-Term Agent Memory, LLM APIs, Temporal, Kubernetes, Docker, Grafana, OAuth2, RBAC